

Name: _____

Date: _____

Period: _____

Density Dry Lab

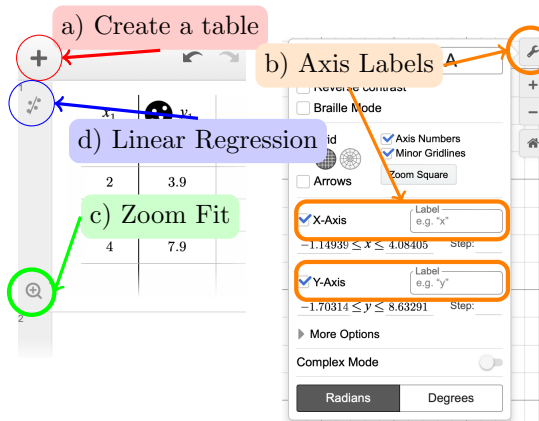
Three students added different volumes of a liquid to a container. Each student used a different liquid (honey, rubbing alcohol, or cooking oil). They then measured the mass of the liquid. They all did this in the same room with the air conditioner turned on. Their results are below:

Volume (mL)	Mass of Honey (g)	Mass of Rubbing Alcohol (g)	Mass of Cooking Oil (g)
10	9.9	8.5	5.4
20	24.0	12.7	20.1
30	40.0	19.8	23.6
40	54.3	35.0	32.5
50	74.8	34.6	50.8
60	81.6	46.9	51.7

1. Your instructor will tell you which set of data to graph. Write it here: _____

2. Go to www.desmos.com/calculator to graph your data.

- Start by making a table by clicking the “+” icon at the top left. You will need to create two separate tables.
- Make sure to label the axes using the wrench icon at the right.
- Zoom out so that you can see the whole graph and so that it fills the page. You can do this using the “Zoom Fit” option, but be careful that your fit does not cut off one of the graphs.
- Create best fit lines for each graph using the “Linear Regression” tool.
- Copy a link to your graph using the export button and clicking “Share a Snapshot”. Paste this link on the appropriate place in Schoology and write it below.



3. Write down the equations for each of the best fit lines below.

You should also copy the link to your graph, paste it on Schoology, and include the link here.

Best Fit Line:

Link to graph: <https://www.desmos.com/calculator/>_____

4. What is the physical meaning of the slope of your graph?

Name:

Date:

Period:

5. The expected values of the densities of each of these substances are as follows: 1.42 g/mL for honey, 0.79 g/mL for alcohol, and 0.92 g/mL for oil. Calculate the percent error of your measured density (that is, slope). Make sure to show your work.

$$\% \text{ error} = \frac{|\text{measured} - \text{expected}|}{\text{expected}} \times 100$$

6. Are your data accurate? Explain.

7. Are your data precise? Explain.